Portugal's Public-Debt Interest Burden: An Empirical Assessment of Measurement, Debt Dynamics, and Euro-Area Comparison

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Abstract

This paper provides a reproducible empirical assessment of Portugal's general-government interest burden from 1960 to 2025, with the main inferential sample restricted to the harmonised Eurostat ESA 2010 period from 1995 to 2025. The dependent variable is interest payable by the general-government sector as a percentage of nominal GDP. The analysis separates three quantities that are frequently conflated in fiscal commentary: the national-accounts interest expense, the interest burden relative to GDP, and the market yield on new benchmark debt. It combines source-provenance review, national-accounts measurement, descriptive periodisation, discrete debt-dynamics accounting, cross-country Eurostat comparison, and deterministic refinancing simulations. The central empirical result is that Portugal's interest burden fell from 5.50 percent of GDP in 1995 to 1.90 percent of GDP in 2025, despite a higher debt ratio in 2025 than in the mid-1990s and despite a material nominal interest bill of EUR 5.96 billion. The sovereign-debt crisis and adjustment period, 2011–2014, remains the highest-burden regime in the harmonised sample, with mean interest expenditure of 4.67 percent of GDP. The post-2015 decline is best understood as the joint result of lower effective interest rates, debt refinancing, nominal GDP growth, and later debt-ratio reduction. By 2025, Portugal ranked fourth among seven selected non-aggregate European comparator countries by interest expenditure as a percentage of GDP, below Italy, Greece, and Spain and above Germany, the Netherlands, and Ireland. The paper emphasises that rate shocks pass through gradually to the average cost of the outstanding stock; the static full-pass-through effect of a 100 basis-point shock at the 2025 debt ratio is approximately 0.90 percentage points of GDP, but this is an arithmetic long-run sensitivity rather than a one-year forecast.

Keywords: Portugal; public debt; interest expenditure; fiscal sustainability; debt dynamics; Eurostat; ESA 2010; euro area.
JEL codes: E62; H62; H63; H68.

Contents

1 Introduction

Interest expenditure is the fiscal price of previously accumulated public liabilities. It is not merely an accounting residual: it constrains the composition of public expenditure, affects the primary balance required to stabilise debt, and mediates the transmission of monetary and sovereign-risk conditions to the public budget. In high-debt economies, a modest movement in the effective interest rate can have large fiscal implications when compounded over the debt stock. Conversely, a high nominal interest bill may coexist with a falling interest burden when nominal GDP grows or when the debt ratio falls.

Portugal is a useful case for studying this interaction. The country entered monetary union after a period of declining long-term yields, experienced a large rise in debt and financing stress during the global financial crisis and the euro-area sovereign-debt crisis, then benefited from lower effective financing costs and later from rapid post-pandemic nominal GDP growth. These episodes make it possible to observe several regimes in one coherent annual dataset: euro convergence, early monetary union, crisis, adjustment, the ECB intervention and refinancing period, the pandemic, and the recent inflation and monetary-tightening regime.

The purpose of this paper is not to estimate a structural fiscal reaction function or to forecast Portugal's debt path. Its purpose is narrower and more foundational: to measure the interest burden correctly, document its evolution, and show how the burden should be interpreted in the presence of changing GDP, debt ratios, effective interest rates, and refinancing lags. The empirical strategy is therefore accounting-based and transparent. It treats national accounts as the measurement system, then uses deterministic identities and carefully labelled comparisons to organise the evidence.

The paper answers seven research questions:

1. What is the annual level of Portugal's general-government interest payable, and how large is it relative to GDP?
2. How different are the euro amount of interest and the GDP-normalised burden as empirical objects?
3. What roles do the debt ratio, the effective interest rate, nominal GDP growth, and the primary balance play in the observed dynamics?
4. How did the burden vary across macro-fiscal regimes from 1995 to 2025?
5. How does Portugal compare with selected European peers under harmonised Eurostat definitions?
6. What is the arithmetic exposure of the burden to benchmark rate shocks?
7. What are the principal measurement and interpretation limits?

The central finding is that Portugal's public-debt interest burden declined substantially over the harmonised Eurostat period. The burden was 5.50 percent of GDP in 1995, 4.80 percent in 2014, 2.90 percent in 2019, and 1.90 percent in 2025. This does not mean that interest payments became fiscally irrelevant. In 2025 the general government still paid EUR 5.96 billion in interest. Rather, the result means that the economy's nominal scale and the effective cost of the outstanding debt stock changed enough to reduce the interest-to-GDP ratio.

2 Conceptual Framework and Measurement

Public discussion often uses the word "interest" to refer to several different objects. This paper distinguishes them explicitly.

First, *interest payable* is a national-accounts flow. In the Eurostat ESA framework it is the interest expense recorded for the general-government sector, not only the central government and not only cash paid in a particular budget line. This is the numerator used in the headline outcome and follows the ESA national-accounts measurement framework [2, 3].

Second, the *interest burden* is interest payable divided by nominal GDP. This is the main dependent variable because it expresses the fiscal flow in relation to the annual income base. The ratio is appropriate for fiscal-space questions because the same nominal payment can be more or less burdensome depending on nominal GDP.

Third, the *market yield* is a price on new or benchmark borrowing. It is not the average cost of the outstanding debt portfolio. Because governments roll over debt gradually, benchmark yields affect the interest bill with a lag whose length depends on maturity, issuance strategy, cash buffers, and the composition of fixed-rate and variable-rate liabilities.

The empirical design of the paper follows from this distinction. The baseline outcome is not the bond yield, and the yield is not treated as if it repriced the entire stock immediately. Instead, the paper measures the national-accounts interest burden and separately analyses the gap between the benchmark yield and the implicit average rate.

Let D_t denote the nominal stock of general-government gross debt at the end of year t, Y_t nominal GDP, I_t interest payable, and PB_t the primary balance. Define $d_t = D_t/Y_t$, $i_t = I_t/Y_t$, and $pb_t = PB_t/Y_t$. A standard discrete debt-dynamics identity writes the change in the debt ratio as:

$$\Delta d_t = \frac{r_t - g_t}{1 + g_t} d_{t-1} - pb_t + sfa_t, \tag{1}$$

where r_t is the effective interest rate on the debt stock, g_t is nominal GDP growth, and sfa_t is the stock-flow adjustment. The first term captures the automatic debt dynamics generated by the difference between the effective interest rate and nominal growth. A positive primary balance reduces the debt ratio, while a positive stock-flow adjustment increases it.

Equation 1 is not a behavioural model. It is an accounting identity once sfa_t is defined as the reconciliation residual. Its value in this paper is diagnostic: it separates periods in which debt pressure comes from the interest-growth differential from periods in which the primary balance or residual stock-flow movements dominate the observed debt change. This accounting tradition is standard in fiscal-sustainability analysis [4, 5].

Ignoring stock-flow adjustments, the primary balance that stabilises the debt ratio is approximately:

$$pb_t^* = \frac{r_t - g_t}{1 + g_t} d_{t-1}. \tag{2}$$

When nominal growth exceeds the effective interest rate, this term can be negative: the debt ratio may fall even with a primary deficit, provided the deficit is not too large and residual stock-flow movements do not offset the effect. When the effective interest rate exceeds nominal growth, a primary surplus is required to stabilise the debt ratio absent stock-flow adjustments.

This stabilisation condition is crucial for interpreting Portugal after 2021. High nominal growth pushed the interest-growth differential into strongly negative territory in 2022 and 2023, lowering the debt-stabilising primary balance. That favourable arithmetic did not eliminate fiscal risk, but it did reduce the near-term debt-ratio pressure from the existing stock.

3 Data and Research Design

The unit of observation is the calendar year. The institutional sector is general government as defined in the European national-accounts framework. This sector choice is important because central-government cash measures can differ from general-government accrual-based national accounts. The paper therefore does not mix central-government budget execution with ESA general-government interest payable.

The main empirical sample is 1995–2025, the period in which the authoritative Portugal series is built from Eurostat ESA 2010 concepts. A linked AMECO extension is preserved for 1960–1994. The extension is useful for historical context, but it is not used to make claims about accounting-consistent changes across the 1995 boundary. The first ESA 2010 main-series year is marked as a basis break.

The core evidence is drawn from Eurostat government-finance statistics, government-debt statistics, national accounts, and the harmonised long-term government bond yield series [3]. These sources provide the four empirical objects required by the paper: interest payable, gross public debt, nominal and real GDP, and a market-yield benchmark. The analysis uses million-euro values when the level of the interest bill is the object of interest and percentage-of-GDP measures when the fiscal burden is the object of interest.

The European comparison is constructed under the same conceptual definitions: general government, annual frequency, harmonised national-accounts concepts, and comparable units. Aggregate geographies, such as the euro area, are retained as contextual information where available but excluded from country ranks. This design prevents the country comparison from being driven by inconsistent accounting definitions.

AMECO is used for a linked pre-1995 extension [1]. The extension is kept separate from the Eurostat main sample and is explicitly labelled as a linked historical series across accounting regimes. It cannot overwrite Eurostat observations. Forecast observations are excluded from historical descriptive statements unless explicitly stated.

This conservative treatment is appropriate because a long time series can create a false appearance of precision when accounting regimes change. The paper therefore uses the extension as context and the Eurostat ESA 2010 sample as the primary evidence base.

The principal burden measure is:

$$b_t = \frac{I_t}{Y_t} \times 100, \tag{3}$$

where I_t is general-government interest payable and Y_t is nominal GDP. The preferred empirical measure is Eurostat's official percentage-of-GDP ratio. A reconstructed ratio from the underlying million-euro values is used only as a reconciliation check and fallback.

The implicit interest rate is:

$$r_t = \frac{I_t}{(D_{t-1} + D_t)/2} \times 100. \tag{4}$$

The denominator is the average of prior-year and current-year debt. This choice reduces mechanical distortion when year-end debt moves sharply. It also matches the paper's emphasis on the average cost of the outstanding stock rather than the marginal cost of new borrowing.

Interest expenditure is recorded as a positive expense. Therefore:

$$pb_t = ob_t + b_t, \tag{5}$$

where ob_t is the overall balance as a percentage of GDP. A government can therefore have an overall deficit and a primary surplus if the deficit is smaller than interest expenditure.

Nominal GDP growth is calculated from current-price GDP. Real GDP growth is read from the Eurostat chain-linked volume percentage-change series. GDP deflator growth is derived as:

$$1 + \pi_t^{GDP} = \frac{1 + g_t^{nominal}}{1 + g_t^{real}}. \tag{6}$$

This multiplicative definition is preferable to subtracting real growth from nominal growth when growth rates are large, as in 2020–2023.

The full-pass-through interest-burden effect of a rate shock is:

$$\Delta b = d\Delta r. \tag{7}$$

At a debt ratio of 89.70 percent of GDP, a 100 basis-point rate shock applied to the full stock implies an additional burden of approximately 0.897 percentage points of GDP. This is a static sensitivity. It should not be interpreted as an immediate forecast because only a fraction of the debt stock is refinanced in a typical year.

3.1 Data Quality and Validation

The empirical series passed the project's data-validation checks. One warning remains: the official and reconstructed debt ratios differ by more than the configured tolerance in 1997 and 1998. The warning is analytically plausible because early years around accounting-basis transitions and official ratio conventions can create small reconciliation differences. It is retained in the paper because a high-quality empirical study should report data warnings rather than hide them.

The analysis depends on three forms of data integrity. First, every observation must represent one and only one calendar year. Second, denominators such as GDP and debt must be positive, finite, and economically interpretable. Third, cross-country comparisons must use the same institutional sector, transaction concept, and unit. The validation layer checks these conditions before the figures and tables in the paper are produced. Technical replication details are reported in the appendices rather than in the main argument.

4 Empirical Results

4.1 Long-Run Portuguese Evidence

The harmonised ESA 2010 sample contains 31 annual observations from 1995 to 2025. Table 1 summarises the main variables. Over this period, interest expenditure averaged 3.29 percent of GDP, with a minimum of 1.90 percent and a maximum of 5.50 percent. The nominal interest bill averaged EUR 5.53 billion, with a maximum of EUR 8.33 billion in 2014. The debt ratio averaged 92.26 percent of GDP, with a minimum of 54.20 percent in 2000 and a maximum of 134.10 percent in 2020.

Table 1: Summary statistics, Portugal ESA 2010 sample, 1995–2025

Variable	Mean	Std. dev.	Min	Min year	Max	Max year
Interest/GDP (%)	3.287	0.969	1.900	2022	5.500	1995
Interest (EUR m)	5531.365	1535.291	3475.400	1998	8334.400	2014
Debt/GDP (%)	92.258	29.324	54.200	2000	134.100	2020
Implicit interest rate (%)	3.849	1.470	1.708	2022	7.934	1996
Overall balance/GDP (%)	-4.035	2.957	-11.400	2010	1.100	2023
Primary balance/GDP (%)	-0.748	2.725	-8.500	2010	3.200	2023
Nominal GDP growth (%)	4.211	4.063	-6.274	2020	12.686	2022
Real GDP growth (%)	1.527	3.021	-8.200	2020	7.000	2022
GDP deflator growth (%)	2.625	1.625	-0.325	2012	7.488	2023
Ten-year yield (%)	4.524	2.718	0.300	2021	11.470	1995

Notes: Observed Eurostat ESA 2010 Portugal rows only. The implicit rate uses average debt. Interest is general-government interest payable.

The most important descriptive fact is the decline in the interest burden. In 1995, Portugal's general government paid interest equal to 5.50 percent of GDP. In 2025 the burden was 1.90 percent of GDP. The change is a reduction of 3.60 percentage points of GDP. In nominal terms, however, the interest bill increased from EUR 5.04 billion to EUR 5.96 billion. This divergence is not a contradiction. It shows why the GDP denominator is essential for fiscal interpretation.

Figure 1 shows the long decline, the crisis-period reversal, and the renewed compression after 2014. The burden fell sharply from the mid-1990s into the early euro period, then rose during the sovereign-debt crisis. It declined again during the ECB intervention and refinancing period and remained near 2 percent of GDP in the inflation and monetary-tightening period.

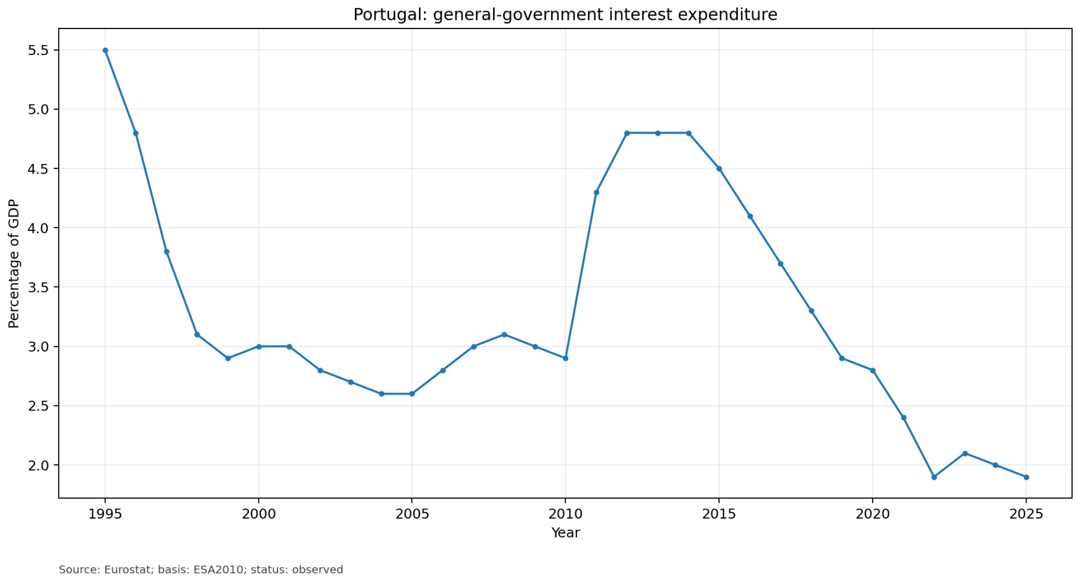

Figure 1: Portugal: interest expenditure as a percentage of GDP.

Figure 2 displays the same interest flow in nominal euros. The euro amount peaked in 2014 at EUR 8.33 billion and then declined through 2022 before increasing again in 2023–2025. The contrast between Figures 1 and 2 is central: a rising nominal bill can coexist with a stable or falling GDP burden when nominal GDP rises rapidly or when the debt ratio falls.

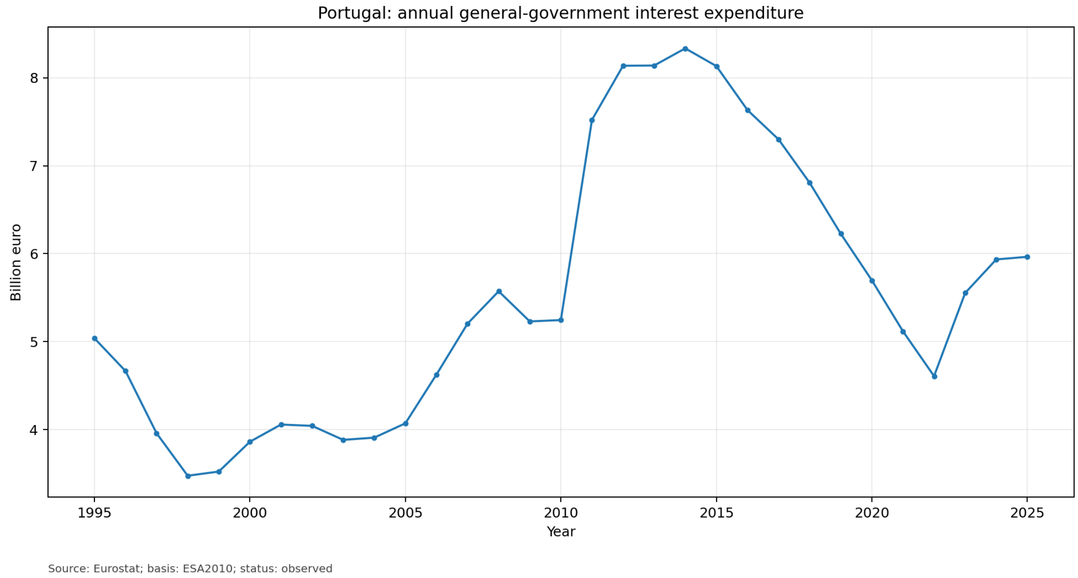

Figure 2: Portugal: interest expenditure in billion euros.

Figure 3 shows the debt ratio and implicit interest rate. The debt ratio fell to 54.20 percent of GDP in 2000, rose to 132.50 percent by 2014, peaked at 134.10 percent in 2020, and then declined to 89.70 percent in 2025. The implicit interest rate moved in the opposite direction over much of the sample: it was very high in the 1990s, declined during monetary union, continued falling after the crisis, reached 1.71 percent in 2022, and was 2.18 percent in 2025.

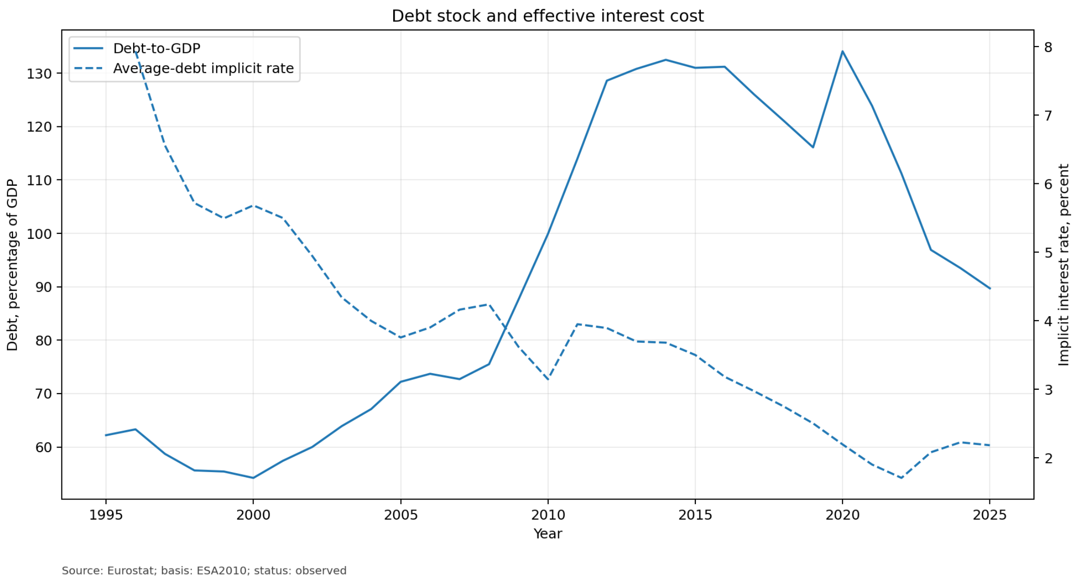

Figure 3: Portugal: debt ratio and implicit interest rate.

The paper does not use a level-correlation matrix as evidence about the drivers of the interest burden. In this setting the main variables are mechanically linked, persistent, and share GDP denominators, so pairwise correlations would mix accounting identities with common trends. The empirical interpretation therefore relies on explicit accounting decompositions: interest expenditure is read as the product of an average-debt financing cost and an average-debt ratio, while debt-ratio changes are decomposed into the interest-growth term, primary balance contribution, and stock-flow residual.

Figure 4 shows the overall balance and the primary balance. Portugal recorded large overall deficits during the global financial crisis, the sovereign-debt crisis, and the pandemic. The primary balance is less negative because it adds back interest expenditure. The difference between the two lines is precisely the interest burden. In 2025 the overall balance was 0.70 percent of GDP and the primary balance was 2.60 percent of GDP.

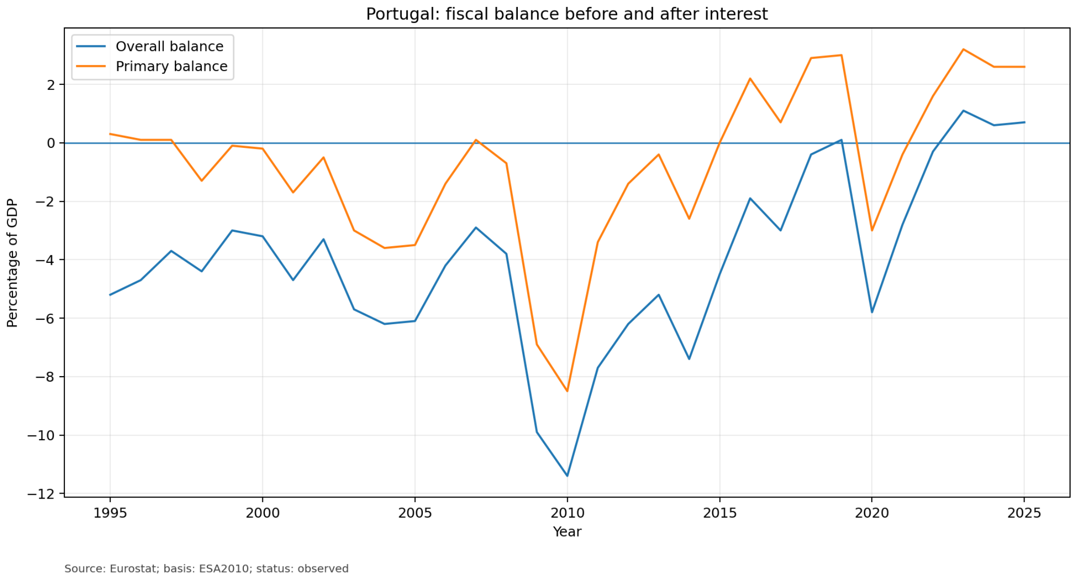

Figure 4: Portugal: overall and primary balances.

Figure 5 compares Portugal's ten-year benchmark yield with the implicit interest rate. The two series are related but not identical. Benchmark yields moved sharply during market-stress episodes, while the implicit rate changed more gradually because it reflects the average cost of the outstanding stock.

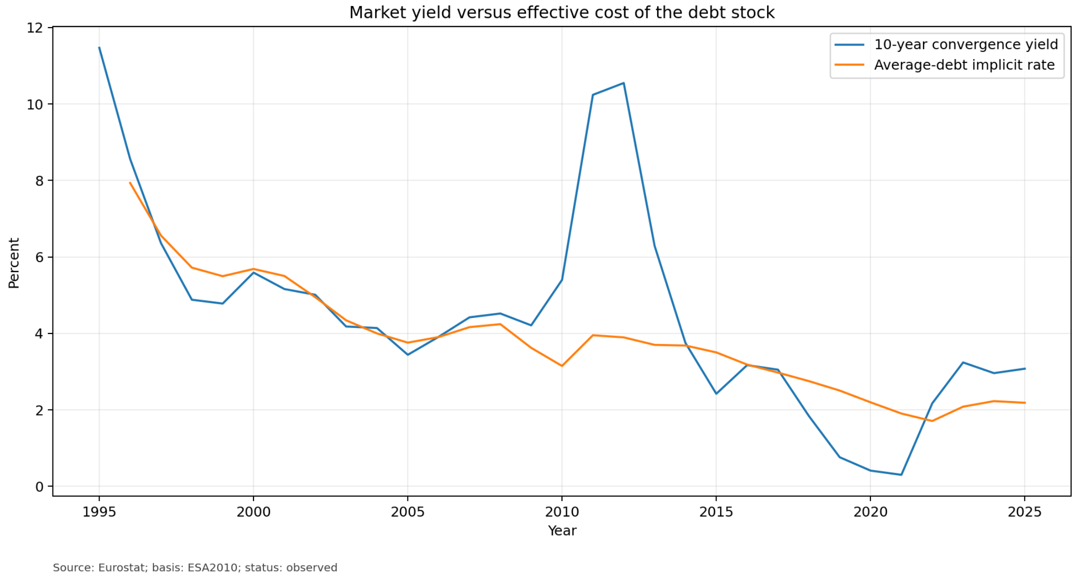

Figure 5: Portugal: benchmark market yield versus implicit interest rate.

This distinction is especially important after the monetary-policy tightening episode. Portugal's ten-year yield rose from exceptionally low levels in 2020–2021 to 3.08 percent in 2025, but the implicit rate in 2025 was 2.18 percent. The budgetary effect of higher yields depends on refinancing, not on instant repricing of the entire stock.

4.2 Regimes and Debt Dynamics

The paper uses descriptive historical regimes rather than formal structural break estimation. The regimes are not causal claims; they are a disciplined way to organise the time series around well-known macro-fiscal episodes. Table 2 reports regime averages.

Table 2: Regime averages, Portugal ESA 2010 sample

Regime	Years	Int./GDP	Int. EUR m	Debt/GDP	Impl. rate	Nom. growth	Prim. bal.
Euro convergence	1995–1998	4.30	4284.68	59.95	6.74	6.74	-0.20
Early euro period	1999–2007	2.82	4131.03	64.07	4.64	5.27	-1.54
Global financial crisis	2008–2010	3.00	5349.47	87.67	3.67	0.85	-5.37
Sovereign-debt crisis and adjustment	2011–2014	4.67	8033.05	126.47	3.81	-0.91	-1.95
ECB intervention and refinancing	2015–2019	3.70	7219.64	125.08	2.98	4.37	1.76
Pandemic	2020–2021	2.60	5407.45	129.00	2.05	0.71	-1.70
Inflation and monetary tightening	2022–2025	1.98	5515.15	97.83	2.05	9.14	2.50

Notes: All entries are regime means. Interest in euros is reported in million euros. The ten-year yield is excluded from the printed table to keep the regime table readable; its regime means are discussed in text.

The convergence period combines a high interest burden with a declining effective rate. The mean interest burden was 4.30 percent of GDP, and the mean implicit interest rate was 6.74 percent. The ten-year yield averaged 7.82 percent. The debt ratio was relatively modest by later Portuguese standards, averaging 59.95 percent of GDP, but the high effective rate made interest expenditure large relative to GDP.

The early euro period had a lower interest burden, averaging 2.82 percent of GDP. The implicit rate fell to a mean of 4.64 percent, while nominal growth averaged 5.27 percent. The debt ratio increased from the low point of 2000 to 72.70 percent by 2007, but the interest burden remained near 3 percent of GDP. This is the first major example in the sample of debt and interest burden moving differently.

During 2008–2010, nominal growth slowed sharply and the primary balance deteriorated. The mean primary balance was -5.37 percent of GDP. The debt ratio rose rapidly, and the overall balance reached its sample minimum of -11.40 percent of GDP in 2010. The interest burden had not yet fully adjusted to later sovereign stress, remaining around 3 percent of GDP on average.

The sovereign-debt crisis and adjustment period is the highest-burden regime in the harmonised sample. Interest expenditure averaged 4.67 percent of GDP and EUR 8.03 billion. The debt ratio averaged 126.47 percent of GDP. Nominal growth was negative on average, and the interest-growth differential was therefore unfavourable. The maximum nominal interest bill, EUR 8.33 billion, occurred in 2014.

Between 2015 and 2019, the interest burden declined from 4.50 percent to 2.90 percent of GDP. The implicit interest rate declined from 3.50 percent to 2.50 percent. Primary balances became positive on average. Debt remained high, but the refinancing environment and effective-rate compression reduced the budgetary burden.

The pandemic period illustrates denominator and stock effects. Real GDP fell 8.20 percent in 2020, nominal GDP fell 6.27 percent, and the debt ratio jumped to 134.10 percent of GDP. Yet the interest burden did not rise; it was 2.80 percent in 2020 and 2.40 percent in 2021. The reason is that the implicit interest rate continued to decline, reaching 1.90 percent in 2021.

The recent period combines higher benchmark yields with strong nominal growth and a declining debt ratio. The interest burden averaged 1.98 percent of GDP. The implicit rate averaged 2.05 percent, while nominal GDP growth averaged 9.14 percent. The debt ratio fell from 111.20 percent in 2022 to 89.70 percent in 2025. This period is not risk-free: nominal interest payments rose by EUR 1.36 billion from 2022 to 2025. But the interest-to-GDP ratio stayed at approximately 2 percent because nominal GDP and the debt ratio moved favourably.

Figure 6 decomposes nominal GDP growth into real growth and GDP deflator growth. The recent period is dominated by strong nominal growth. In 2022 nominal GDP grew 12.69 percent, real GDP grew 7.00 percent, and GDP deflator growth was 5.31 percent. In 2023 nominal GDP grew 10.82 percent and deflator growth was 7.49 percent. These values materially reduced debt-ratio pressure because the denominator grew quickly.

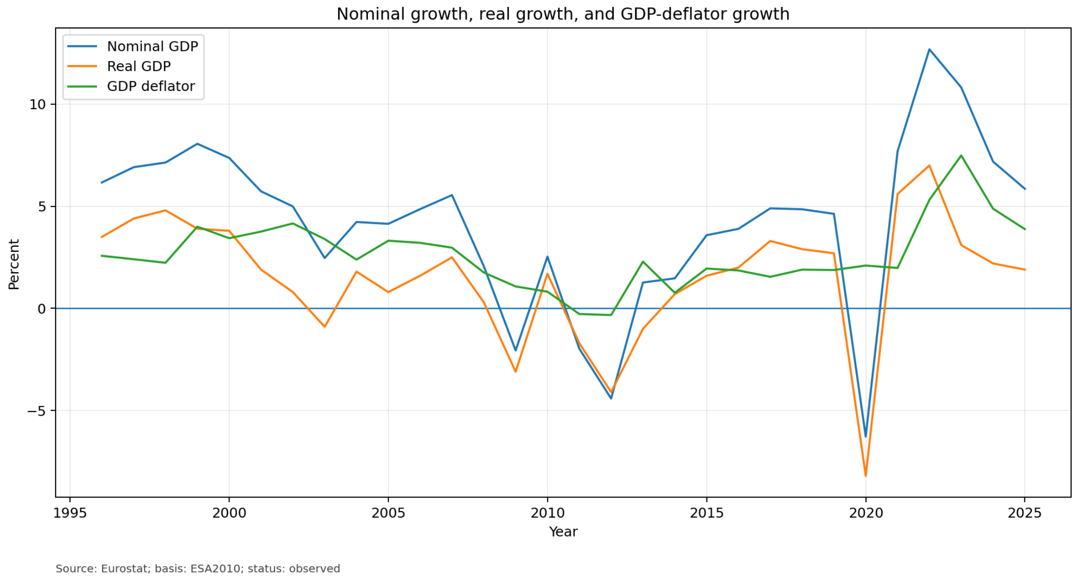

Figure 6: Portugal: nominal growth, real growth, and GDP-deflator growth.

Figure 7 reports the accounting decomposition implied by Equation 1. The pandemic year 2020 is the clearest example of adverse debt dynamics: nominal GDP contracted sharply, pushing the interest-growth term to 10.49 percentage points of GDP. By contrast, 2022 and 2023 show strongly favourable interest-growth arithmetic. The debt-stabilising primary balance was -12.07 percent of GDP in 2022 and -8.77 percent in 2023, before stock-flow effects.

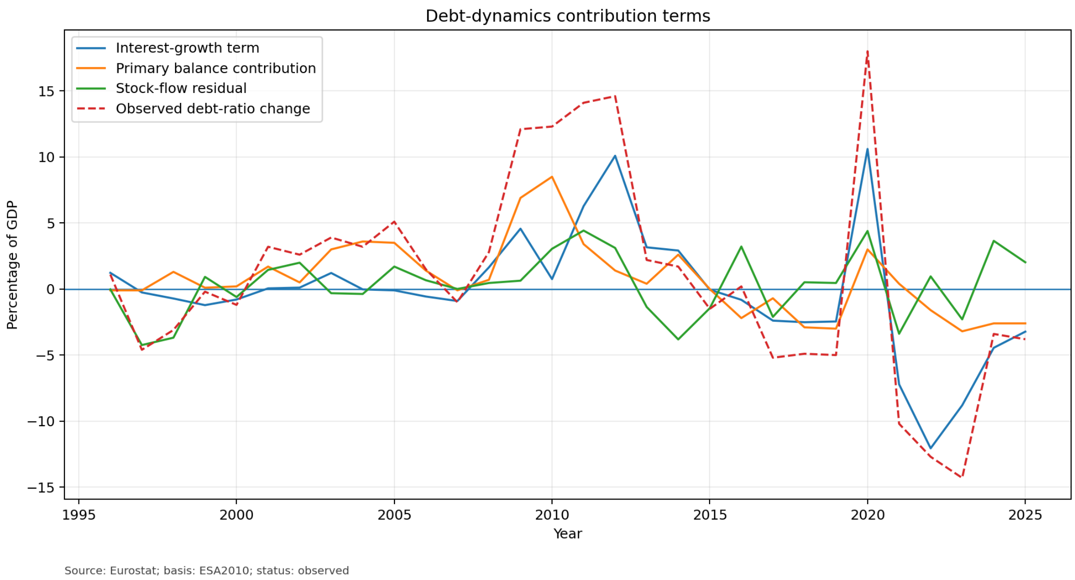

Figure 7: Portugal: debt-dynamics decomposition.

The stock-flow adjustment is a residual. It absorbs valuation effects, financial transactions, cash-debt timing differences, and any discrepancy between the simple debt-dynamics identity and the measured debt ratio. In this dataset, it ranges from -4.29 percentage points of GDP in 1997 to 4.67 percentage points in 2011. It should not be interpreted as an estimated policy instrument. It is an accounting reconciliation term.

Table 3 provides recent annual observations. The key pattern is visible in the last four rows.

From 2022 to 2025 the nominal interest bill rose from EUR 4.61 billion to EUR 5.96 billion, but interest as a share of GDP remained at 1.90 percent in both endpoint years. Over the same period the debt ratio fell by 21.50 percentage points, while the implicit interest rate rose by 0.48 percentage points.

Table 3: Recent annual dynamics, Portugal

Year	Int./GDP (%)	Interest (EUR m)	Debt/GDP (%)	Impl. rate (%)	Overall bal. (% GDP)	Primary bal. (% GDP)	Nom. growth (%)
2014	4.8	8334.4	132.5	3.68	-7.4	-2.6	1.47
2015	4.5	8132.8	131.0	3.50	-4.5	0.0	3.58
2016	4.1	7633.3	131.2	3.18	-1.9	2.2	3.90
2017	3.7	7299.4	126.0	2.97	-3.0	0.7	4.90
2018	3.3	6805.8	121.1	2.75	-0.4	2.9	4.85
2019	2.9	6226.9	116.1	2.50	0.1	3.0	4.63
2020	2.8	5697.1	134.1	2.20	-5.8	-3.0	-6.27
2021	2.4	5117.8	123.9	1.90	-2.8	-0.4	7.69
2022	1.9	4608.0	111.2	1.71	-0.3	1.6	12.69
2023	2.1	5553.3	96.9	2.08	1.1	3.2	10.82
2024	2.0	5934.8	93.5	2.23	0.6	2.6	7.19
2025	1.9	5964.5	89.7	2.18	0.7	2.6	5.85

4.3 European Comparison and Rate-Shock Exposure

The comparator panel includes Portugal, Spain, Italy, Greece, Ireland, Germany, the Netherlands, and euro-area aggregate context where available. The country ranking excludes aggregate geographies. Each country-year observation uses the same Eurostat concepts for interest, GDP, debt, balance, and real growth. This prevents a common error in fiscal comparison: mixing country-specific cash or budgetary definitions with national-accounts definitions.

Table 4 reports the 2025 country ranking by interest expenditure as a percentage of GDP. Portugal ranked fourth among the seven selected non-aggregate countries. Its burden was below Italy, Greece, and Spain, but above Germany, the Netherlands, and Ireland.

Table 4: European comparator panel, 2025

Rank	Country	Interest/GDP (%)	Debt/GDP (%)	Impl. rate (%)	Ten-year yield (%)	Primary bal. (% GDP)
1	Italy	3.9	137.1	2.87	3.59	0.8
2	Greece	3.2	146.1	2.15	3.37	4.9
3	Spain	2.4	100.7	2.43	3.21	0.0
4	Portugal	1.9	89.7	2.18	3.08	2.6
5	Germany	1.1	63.5	1.79	2.59	-1.6
6	Netherlands	0.7	44.4	1.67	2.80	-0.9
7	Ireland	0.5	32.9	1.40	2.89	2.3

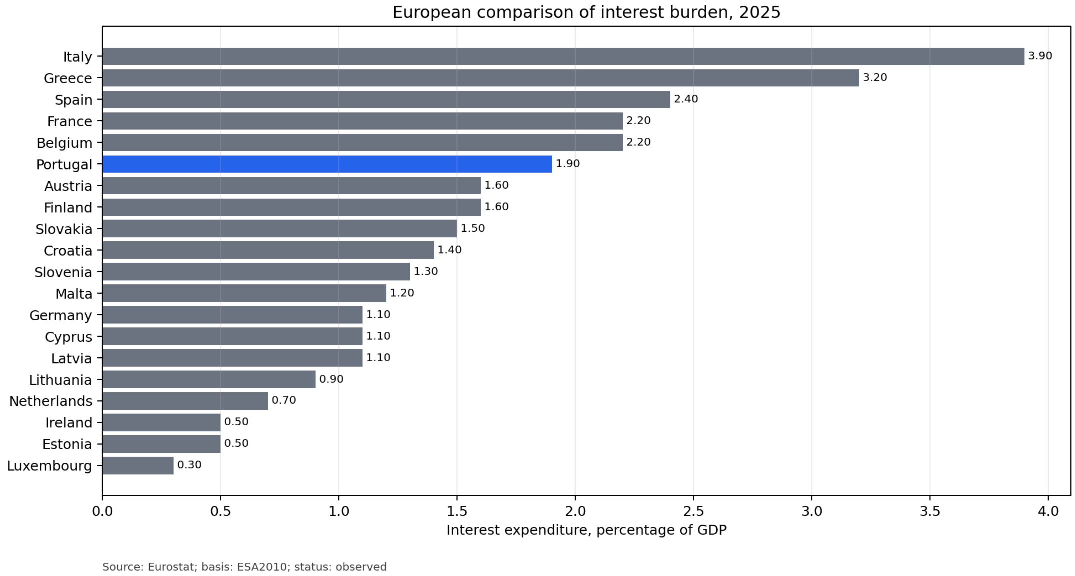

Figure 8: European comparison of interest expenditure as a percentage of GDP.

Portugal's intermediate position is consistent with three facts. First, Portugal's debt ratio remained high by northern euro-area standards but was below Italy, Greece, and Spain in 2025. Second, Portugal's implicit interest rate was moderate relative to Spain and Italy. Third, Portugal ran a sizeable primary surplus in 2025, which supported debt reduction. The comparison therefore does not show that Portugal is low-risk in an absolute sense; it shows that its interest burden was no longer among the highest in the selected country set under harmonised definitions.

Table 5 reports static sensitivities at the 2025 debt ratio. A 50 basis-point full-stock shock implies 0.449 percentage points of GDP in additional annual interest. A 100 basis-point shock implies 0.897 percentage points. A 200 basis-point shock implies 1.794 percentage points. These numbers are useful for scale, but they deliberately overstate first-year budgetary effects when most debt is fixed-rate and matures gradually.

Table 5: Static full-pass-through sensitivities at 2025 debt ratio

Debt/GDP (%)	Shock (bps)	Shock rate decimal	Additional interest/GDP (percentage points)
89.70	50	0.005	0.449
89.70	100	0.010	0.897
89.70	200	0.020	1.794

Figure 9 reports deterministic refinancing paths using the configured refinancing shares. The purpose is not to forecast Portugal's future debt-servicing cost. The purpose is to show that the arithmetic effect of rate shocks is distributed over time when only part of the stock is repriced each year.

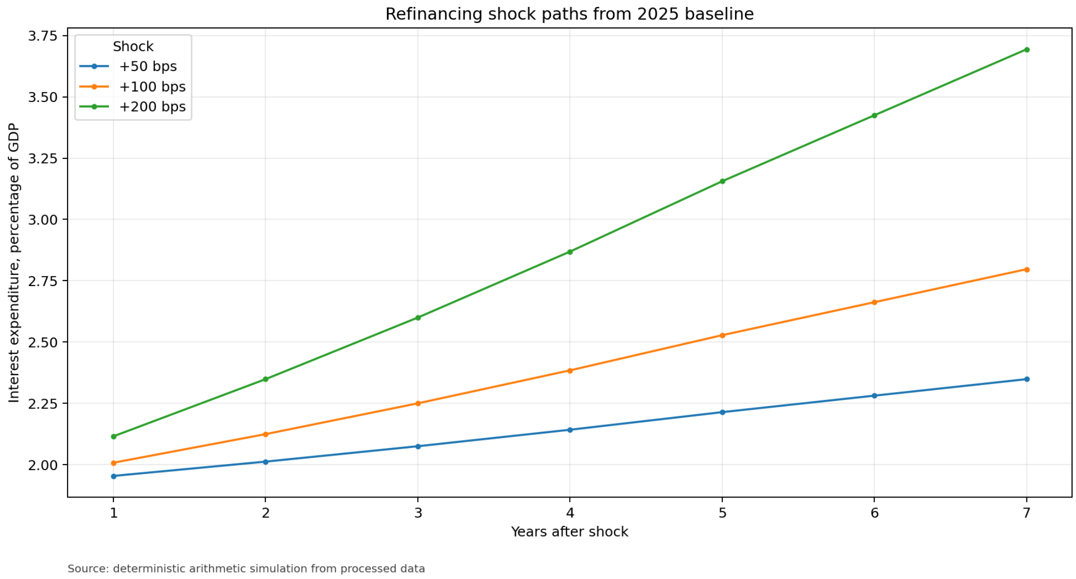

Figure 9: Gradual refinancing shock paths.

The refinancing exercise has two fiscal implications. First, maturity structure is a form of risk management because it determines how quickly market conditions enter the budget. Second, favourable recent debt arithmetic should not be extrapolated mechanically. If higher market yields persist, the average rate can rise gradually even without an immediate fiscal shock.

5 Discussion and Limitations

The evidence indicates that no single variable explains the full decline in Portugal's interest burden. The long-run decline from 1995 to 2025 reflects:

- a large fall in the effective interest rate from high pre-euro and convergence-period levels;
- monetary-union and later ECB-era financing conditions that lowered the average cost of the outstanding stock;
- nominal GDP growth, especially after the pandemic;
- a large decline in the debt ratio after the 2020 peak;
- improved primary balances in several recent years.

The crisis period shows the opposite combination: high debt, weak or negative nominal growth, and elevated financing stress. The recent period shows why high nominal growth can quickly reduce debt-ratio pressure, but also why interest in euros can rise even when the ratio is stable.

The paper does not prove that Portugal's fiscal position is structurally safe. It does not estimate future growth, future primary balances, or future market risk premia. It also does not model contingent liabilities, ageing costs, or debt-maturity composition in full detail. Instead, it provides the measurement and accounting basis required before those questions can be studied rigorously.

The contribution of this study is primarily empirical and reproducible. It does not propose a new macroeconomic theory. It provides a clean, auditable, annually indexed dataset that separates national-accounts interest flows, debt stocks, growth, balances, market yields, and comparator-country metrics. It also embeds validation checks that reduce the probability of silent data corruption. For policy analysis, this is a necessary foundation: fiscal interpretation is only as strong as the measurement layer underneath it.

Five limitations are material.

First, the pre-1995 AMECO extension is not accounting-equivalent to the Eurostat ESA 2010 main sample. It is labelled and preserved, but the core claims are based on 1995–2025.

Second, ESA interest payable is not the same as central-government cash interest. Users interested in budget execution, Treasury cash management, or debt-office operations would need a separate reconciliation layer.

Third, the implicit interest rate is an average-cost measure, not a marginal cost of funds. It is useful for debt dynamics but cannot replace an issuance yield in debt-management analysis.

Fourth, the cross-country panel is harmonised by Eurostat definitions but does not control for maturity structure, inflation-linked debt, cash buffers, official-sector loans, or country-specific fiscal institutions.

Fifth, rate-shock simulations are deterministic arithmetic exercises. They omit feedback through growth, inflation, primary balances, market risk premia, and policy responses.

6 Conclusion

Portugal's general-government interest burden has declined dramatically in the harmonised ESA 2010 sample. The burden fell from 5.50 percent of GDP in 1995 to 1.90 percent in 2025. The nominal interest bill, however, remained material: EUR 5.96 billion in 2025. This distinction is the core empirical lesson of the paper. The euro amount, the GDP ratio, the debt ratio, the effective interest rate, and the market yield are related but distinct objects.

The sovereign-debt crisis and adjustment period remains the highest-burden regime in the sample. The post-2015 decline reflects a lower average cost of debt, refinancing effects, primary-balance improvement, and later nominal GDP growth and debt-ratio reduction. The recent tightening period has not yet translated one-for-one into the interest burden because the debt stock reprices gradually and because the GDP denominator grew strongly. This does not remove the risk from sustained higher rates; it clarifies the channel through which that risk reaches the budget.

In the 2025 comparator panel, Portugal occupied an intermediate position among selected European countries. It had a lower interest burden than Italy, Greece, and Spain, and a higher burden than Germany, the Netherlands, and Ireland. That ranking should be read as a harmonised descriptive comparison, not as a full fiscal-sustainability judgement.

A Appendix A: Annual ESA 2010 Portugal Table

Table 6: Annual Portugal observations, 1995–2025

Year	Int./GDP (%)	Interest (EUR m)	Debt/GDP (%)	Impl. rate (%)	Overall bal. (% GDP)	Primary bal. (% GDP)	Nom. growth (%)
1995	5.5	5038.9	62.2	–	-5.2	0.3	–
1996	4.8	4666.1	63.3	7.93	-4.7	0.1	6.89
1997	3.9	3890.1	59.5	6.15	-3.7	0.2	7.16
1998	3.0	3475.4	54.8	6.12	-3.2	-0.2	6.16
1999	2.9	3523.0	55.4	5.49	-3.1	-0.2	6.72
2000	3.0	3861.9	54.2	5.69	-3.2	-0.2	6.57
2001	2.9	3969.2	56.2	5.45	-4.8	-1.9	5.47
2002	2.8	4042.2	59.9	4.95	-3.3	-0.5	4.59
2003	2.7	3932.2	63.9	4.46	-4.7	-2.0	1.54
2004	2.7	4102.1	67.5	4.18	-5.8	-3.1	3.52
2005	2.8	4470.5	72.2	4.05	-6.2	-3.4	4.05
2006	2.8	5072.1	74.6	4.05	-4.3	-1.5	5.00
2007	3.0	5205.1	72.7	4.16	-2.9	0.1	4.57
2008	3.0	5361.2	75.6	4.06	-3.7	-0.7	2.27
2009	2.9	5141.6	85.8	3.58	-10.2	-7.3	-1.99
2010	3.1	5545.6	101.6	3.37	-11.4	-8.5	2.27
2011	4.3	7521.6	114.0	3.95	-7.7	-3.4	-1.44
2012	4.8	8075.3	129.0	3.90	-5.8	-1.0	-3.91
2013	4.8	8200.9	130.4	3.70	-4.8	0.0	0.24
2014	4.8	8334.4	132.5	3.68	-7.4	-2.6	1.47
2015	4.5	8132.8	131.0	3.50	-4.5	0.0	3.58
2016	4.1	7633.3	131.2	3.18	-1.9	2.2	3.90
2017	3.7	7299.4	126.0	2.97	-3.0	0.7	4.90
2018	3.3	6805.8	121.1	2.75	-0.4	2.9	4.85
2019	2.9	6226.9	116.1	2.50	0.1	3.0	4.63
2020	2.8	5697.1	134.1	2.20	-5.8	-3.0	-6.27
2021	2.4	5117.8	123.9	1.90	-2.8	-0.4	7.69
2022	1.9	4608.0	111.2	1.71	-0.3	1.6	12.69
2023	2.1	5553.3	96.9	2.08	1.1	3.2	10.82
2024	2.0	5934.8	93.5	2.23	0.6	2.6	7.19
2025	1.9	5964.5	89.7	2.18	0.7	2.6	5.85

B Appendix B: Variable Definitions

Table 7: Core analytical variables

Column	Unit	Definition
`year`	year	Calendar year.
`interest_mio_eur`	EUR million	General-government interest payable.
`interest_pct_gdp`	percent of GDP	Preferred interest burden: official Eurostat ratio with calculated fallback.
`nominal_gdp_mio_eur`	EUR million	GDP at current market prices.
`debt_mio_eur`	EUR million	Maastricht consolidated gross debt.
`debt_pct_gdp`	percent of GDP	Preferred debt ratio.
`implicit_interest_rate_pct`	percent	Interest payable divided by configured debt denominator.
`overall_balance_pct_gdp`	percent of GDP	Net lending or borrowing.
`primary_balance_pct_gdp`	percent of GDP	Overall balance plus interest expenditure.
`nominal_gdp_growth_pct`	percent	Annual growth of current-price GDP.
`real_gdp_growth_pct`	percent	Chain-linked volume growth rate.

Column	Unit	Definition
gdp_deflator_growth_pct	percent	Derived GDP deflator growth.
interest_growth_differential_pct	percentage points	Implicit interest rate minus nominal GDP growth.
debt_stabilising_primary_balance_pct_gdp	percent of GDP	Primary balance required to stabilise debt absent stock-flow adjustment.
stock_flow_adjustment_pct_gdp	percent of GDP	Debt-dynamics residual.
ten_year_yield_pct	percent	Long-term government bond yield.
source	category	Source family, Eurostat or AMECO.
accounting_basis	category	Accounting framework label.
observation_status	category	Observed or forecast status.
retrieval_timestamp_utc	timestamp	Joined UTC source retrieval timestamp where available.
source_checksum_sha256	hash	Joined source checksum metadata.

C Appendix C: Source and Validation Notes

Eurostat source tables are `gov_10a_main`, `gov_10dd_edpt1`, `nama_10_gdp`, and `irt_lt_mcby_a`. AMECO is used for the linked historical extension. The latest validation result passed all error-severity checks and retained one warning for debt-ratio reconciliation in 1997 and 1998.

D Appendix D: Replication Materials

The replication workflow produces the following analytical outputs:

- `data/processed/portugal_debt_interest.csv`;
- `data/processed/portugal_debt_interest.sqlite`;
- `data/processed/eurostat_panel_metrics.csv`;
- `reports/validation.json`;
- `reports/summary.md`;
- publication-oriented figures in `reports/figures/`.

Raw source downloads include retrieval timestamps, request metadata, payload sizes, and checksums. Processed rows preserve source-vintage and checksum metadata so revisions can be audited.

E Appendix E: Reproducibility Commands

```
python -m pt_debt_interest.cli all --config config/default.yaml
pytest
ruff check .
mypy src
```

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  portugal_public_debt_interest_report.tex
```

References

[1] European Commission. *AMECO macro-economic database*. Directorate-General for Economic and Financial Affairs.

[2] Eurostat. *European System of Accounts: ESA 2010*. Publications Office of the European Union.

[3] Eurostat. *Government finance statistics: revenue, expenditure, deficit, debt, GDP, and long-term interest rate datasets*. Eurostat database.

[4] Escolano, J. *A Practical Guide to Public Debt Dynamics, Fiscal Sustainability, and Cyclical Adjustment of Budgetary Aggregates*. International Monetary Fund Technical Notes and Manuals.

[5] Blanchard, O. *Public Debt and Low Interest Rates*. American Economic Review.